Highlights from two astronomy problems for Lagrange's equations: the cosmological redshift and the Lagrange points

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November 3, 2026

These notes use only what is in Taylor, *Classical Mechanics*, Sections 7.5, 7.6 and 7.8: Lagrange's equations, a coordinate change that depends on time, ignorable coordinates, and "no t in $\mathcal{L}$ gives a conserved Hamiltonian." The cosmology half follows the standard textbook route for a particle in an expanding universe (P. J. E. Peebles, *The Large-Scale Structure of the Universe*, Princeton, 1980, Chapter II); the second half is the restricted three-body problem as every celestial-mechanics book sets it up (e.g. Murray and Dermott, *Solar System Dynamics*, Cambridge, 1999, Chapter 3). Both halves are Newtonian throughout, and the places where general relativity would add something are flagged.

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1 The redshift: Noether in an expanding universe

The setup. Hubble's law says a galaxy at distance r recedes from us at $v = (\dot{a}/a)\, r$, where $a(t)$ is the *scale factor*: every cosmological distance grows as $r(t) = a(t)\, x$, with x a label that does not change. Coordinates that expand with the universe like this are called *comoving coordinates*: a galaxy that just rides the expansion (the "Hubble flow") sits at a fixed x forever. Differentiating,

$$\dot{r} = \dot{a}\, x + a\, \dot{x}, \tag{1.1}$$

the Hubble flow plus the *peculiar velocity* $v_{\text{pec}} \equiv a\dot{x}$, the motion relative to the flow (a galaxy falling toward a cluster, say, or a photon crossing the universe).

The result, first. For a particle moving freely through a homogeneous expanding universe, in comoving coordinates,

$$\boxed{\mathcal{L} = \tfrac{1}{2}\, m\, a(t)^2\, \dot{x}^2} \qquad \text{(up to a total time derivative, which changes nothing).} \tag{1.2}$$

Two things follow immediately from Taylor's Section 7.8:

- x does not appear in $\mathcal{L}$, so it is ignorable: $p \equiv \partial\mathcal{L}/\partial\dot{x} = ma^2\dot{x}$ is conserved. In terms of the peculiar velocity, $m\, a\, v_{\text{pec}}$ is constant, so

$$v_{\text{pec}} \propto \frac{1}{a}. \tag{1.3}$$

  Random motions die away as the universe grows ("Hubble drag"): when the universe doubles in size, every freely coasting peculiar velocity halves.

- t *does* appear in $\mathcal{L}$, through $a(t)$, so there is no conserved Hamiltonian and no conserved energy. The expanding universe does not have time-translation symmetry, and Noether's theorem has nothing to give.

The rest of this section is where (1.2) comes from, and then what it says about light.

Step 1: Newton's law for the expansion. Fill space with matter of uniform density $\rho(t)$. Pick any point as the origin. A particle at r feels only the gravity of the sphere inside it (Newton's shell theorem; that the outside cancels is the one place Newtonian cosmology has to wave its hands, and general relativity makes the argument honest), so

$$\ddot{r} = -\frac{GM(<r)}{r^2} = -\frac{G}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{4\pi G}{3}\,\rho\, r, \qquad \text{i.e.} \qquad \Phi(r) = \frac{2\pi G}{3}\,\rho\, r^2. \tag{1.4}$$

A particle riding the Hubble flow has $r = ax$ with x constant, so $\ddot{r} = \ddot{a}\,x$ and

$$\ddot{a} = -\frac{4\pi G}{3}\,\rho\, a. \tag{1.5}$$

This is the Friedmann acceleration equation for a universe of pressureless matter. (General relativity's version has extra terms for pressure and for the cosmological constant; with those set to zero it is exactly this.)

Step 2: the Lagrangian in comoving coordinates. Work in one dimension; three is the same with vectors. A particle that is *not* riding the flow has $r = ax$ with $x(t)$ free, and

$$\begin{aligned}\mathcal{L} = \tfrac{1}{2}m\dot{r}^2 - m\Phi &= \tfrac{1}{2}m\,(\dot{a}x + a\dot{x})^2 - \frac{2\pi G}{3}\,m\rho\, a^2x^2 \\ &= \tfrac{1}{2}ma^2\dot{x}^2 \;+\; ma\dot{a}\,x\dot{x} \;+\; \tfrac{1}{2}m\dot{a}^2x^2 \;-\; \frac{2\pi G}{3}\,m\rho\, a^2x^2.\end{aligned} \tag{1.6}$$

The cross term is the one to tame. Since $\frac{d}{dt}\left(\frac{1}{2}ma\dot{a}\,x^2\right) = \frac{1}{2}m(\dot{a}^2 + a\ddot{a})\,x^2 + ma\dot{a}\,x\dot{x}$,

$$ma\dot{a}\,x\dot{x} = \frac{d}{dt}\left(\tfrac{1}{2}ma\dot{a}\,x^2\right) - \tfrac{1}{2}m\dot{a}^2x^2 - \tfrac{1}{2}ma\ddot{a}\,x^2. \tag{1.7}$$

Put that into (1.6). The $\frac{1}{2}m\dot{a}^2x^2$ terms cancel, and by (1.5) $-\frac{1}{2}ma\ddot{a}\,x^2 = +\frac{2\pi G}{3}m\rho a^2x^2$, which cancels the potential term exactly. What is left is

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{dF}{dt}, \qquad F \equiv \tfrac{1}{2}ma\dot{a}\,x^2. \tag{1.8}$$

Step 3: a total derivative in $\mathcal{L}$ changes nothing. The action is $S = \int_{t_1}^{t_2}\mathcal{L}\,dt$, and $\int_{t_1}^{t_2}(dF/dt)\,dt = F(t_2) - F(t_1)$ is fixed by the endpoints: it is the same number for every path between them, so it cannot change which path makes S stationary. (Check: put $\mathcal{L} = dF/dt = \partial_t F + \dot{x}\,\partial_x F$ into Lagrange's equation by itself and everything cancels.) So we may drop it, and (1.2) is the Lagrangian. Had there been lumps in the universe, a galaxy cluster say, (1.2) would pick up $-m\phi(x,t)$ with ϕ the *peculiar* potential of the lumps alone; that is Peebles's starting point for structure formation, and it is why the smooth background's potential vanished here: it is already accounted for by the expansion.

The Hamiltonian, and what is not conserved. With $p = ma^2\dot{x}$,

$$\mathcal{H} = p\dot{x} - \mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}mv_{\text{pec}}^2, \tag{1.9}$$

which is the kinetic energy in the peculiar velocity, and Taylor's general result $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ gives

$$\frac{d\mathcal{H}}{dt} = -ma\dot{a}\,\dot{x}^2 = -2\,\frac{\dot{a}}{a}\,\mathcal{H}, \qquad \text{so} \qquad \mathcal{H} \propto \frac{1}{a^2}, \tag{1.10}$$

consistent with (1.3). The energy is not conserved, and the formalism tells us exactly why: $\partial\mathcal{L}/\partial t \neq 0$.

Light. The argument above assumed $\frac{1}{2}mv^2$, so by itself it says nothing about photons. General relativity makes the same statement for any freely moving particle, massive or not: the momentum measured by comoving observers falls as $1/a$. (The symmetry argument is the one you just saw. The spacetime of a homogeneous universe does not depend on the comoving position x, so the momentum conjugate to x is conserved, and converting it to what a local observer measures costs one factor of a.) For a photon $p = h/\lambda$, so

$$\lambda \propto a, \qquad 1 + z \equiv \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}} = \frac{a_{\text{obs}}}{a_{\text{emit}}}. \tag{1.11}$$

A photon emitted when the universe was half its present size arrives with twice its wavelength: $z = 1$. That is the cosmological redshift, and it is the same conservation law as (1.3).

Where does the photon's energy go? Nowhere. $E = pc \propto 1/a$ simply decreases. The question presumes a conserved total energy, and Noether's theorem says where that would have to come from: time-translation symmetry. An expanding universe does not have it, so there is no global energy to balance the books with. (Locally, in any small freely-falling laboratory, energy is conserved as usual; it is the bookkeeping across cosmological distances that has no ledger.) Compare the slide from Monday: "if $\mathcal{L}$ does not depend explicitly on time, then energy is conserved." This is the contrapositive, with the whole universe as the example.

2 The Lagrange points: Lagrange's equations in a rotating frame

The setup. JWST sits about 1.5×10^6 km from Earth, on the far side from the Sun, at the point called L_2; Jupiter's Trojan asteroids cluster 60° ahead of and behind the planet, at L_4 and L_5. Lagrange found these five points in 1772 as special solutions of the three-body problem (the first Trojan, 588 Achilles, was not seen until 1906). In what sense is a point "at rest" when the Sun, the Earth and the point are all orbiting? In the frame that rotates with the two big bodies. Taylor does rotating frames with vectors in Chapter 9; here the same physics drops out of a Lagrangian in three lines, by the time-dependent coordinate change of Section 7.5.

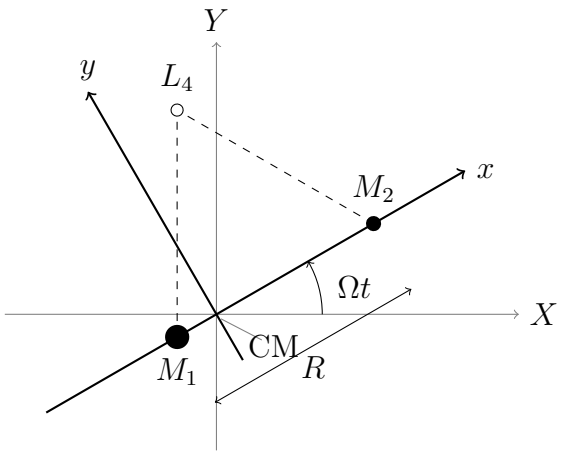
Two bodies M_1 and M_2 move on circular orbits about their center of mass (CM), a distance R apart, at angular speed Ω. Newton for M_1, which is $r_1 = M_2R/(M_1 + M_2)$ from the CM, gives $M_1\Omega^2 r_1 = GM_1M_2/R^2$, so

$$\Omega^2 = \frac{G(M_1 + M_2)}{R^3} \tag{2.1}$$

(Kepler's third law; Chapter 8 does the general case). A third body of mass m, so small that it does not disturb the first two (this is the *restricted* three-body problem), moves in their orbital plane. Write $\mu \equiv M_2/(M_1 + M_2)$, so the CM is at the origin with M_1 at $(-\mu R, 0)$ and M_2 at $((1-\mu)R, 0)$ in coordinates (x, y) that rotate with them:

$$X = x\cos\Omega t - y\sin\Omega t, \qquad Y = x\sin\Omega t + y\cos\Omega t, \tag{2.2}$$

where (X, Y) are inertial.



The result, first.

$$\boxed{\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega\,(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) - U(x, y)} \tag{2.3}$$

with U the ordinary gravitational potential energy of m in the field of the two fixed primaries. Three things follow:

- Lagrange's equations are Newton's second law plus two extra forces, the centrifugal force $m\Omega^2(x, y)$ and the Coriolis force $2m\Omega(\dot{y}, -\dot{x})$: Chapter 9's results, for free.
- A body can sit still in the rotating frame exactly where the *effective potential* $U_{\text{eff}} \equiv U - \tfrac{1}{2}m\Omega^2(x^2 + y^2)$ has zero gradient. There are five such points. Two of them, L_4 and L_5, form equilateral triangles with M_1 and M_2, for any mass ratio; the other three lie on the line through the primaries.
- $\mathcal{L}$ has no explicit t, so $\mathcal{H}$ is conserved. It is $\tfrac{1}{2}mv_{\text{rot}}^2 + U_{\text{eff}}$, called the *Jacobi integral*, and it is *not* $T + U$.

Step 1: the kinetic energy. Differentiate (2.2):

$$\dot{X} = (\dot{x} - \Omega y)\cos\Omega t - (\dot{y} + \Omega x)\sin\Omega t, \qquad \dot{Y} = (\dot{x} - \Omega y)\sin\Omega t + (\dot{y} + \Omega x)\cos\Omega t, \tag{2.4}$$

so the sines and cosines drop out of the speed:

$$\dot{X}^2 + \dot{Y}^2 = (\dot{x} - \Omega y)^2 + (\dot{y} + \Omega x)^2 = \dot{x}^2 + \dot{y}^2 + 2\Omega\,(x\dot{y} - y\dot{x}) + \Omega^2(x^2 + y^2). \tag{2.5}$$

Half of m times this is the kinetic energy in (2.3). This is Taylor's bead on a spinning hoop with two coordinates instead of one: the inertial position depends on the generalized coordinates *and* on t, and that is what breeds the extra terms.

Step 2: the potential energy. In the rotating frame the primaries do not move, so

$$U(x, y) = -\frac{GmM_1}{\sqrt{(x + \mu R)^2 + y^2}} - \frac{GmM_2}{\sqrt{(x - (1 - \mu)R)^2 + y^2}} \tag{2.6}$$

has no t in it. That is the whole reason to rotate: in the inertial frame $U(X, Y, t)$ is time-dependent and nothing is conserved.

Step 3: Lagrange's equations. From (2.3), $\partial\mathcal{L}/\partial\dot{x} = m(\dot{x} - \Omega y)$ and $\partial\mathcal{L}/\partial x = m\Omega\dot{y} + m\Omega^2 x - \partial U/\partial x$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{x} = \partial\mathcal{L}/\partial x$ reads $m\ddot{x} - m\Omega\dot{y} = m\Omega\dot{y} + m\Omega^2 x - \partial U/\partial x$, i.e.

$$m\ddot{x} = -\frac{\partial U}{\partial x} + m\Omega^2 x + 2m\Omega\dot{y}, \qquad m\ddot{y} = -\frac{\partial U}{\partial y} + m\Omega^2 y - 2m\Omega\dot{x}. \tag{2.7}$$

Gravity, then the centrifugal force $m\Omega^2(x, y)$, outward from the axis, then the Coriolis force $2m\Omega(\dot{y}, -\dot{x})$, perpendicular to the velocity. (Chapter 9 writes them as $-m\,\omega \times (\omega \times \mathbf{r})$ and $-2m\,\omega \times \dot{\mathbf{r}}$ with $\omega = \Omega\hat{\mathbf{z}}$; check that those give the same components.) The Coriolis force depends on velocity, so it has no potential energy, but it does no work: it is always perpendicular to $\dot{\mathbf{r}}$.

Step 4: equilibrium. To sit still in the rotating frame, $\dot{x} = \dot{y} = 0$ and $\ddot{x} = \ddot{y} = 0$, so the Coriolis force is off and (2.7) says

$$\boldsymbol{\nabla} U_{\text{eff}} = 0, \qquad U_{\text{eff}}(x, y) \equiv U(x, y) - \tfrac{1}{2}m\Omega^2(x^2 + y^2). \tag{2.8}$$

The centrifugal force came from a potential, so it joins U. Figure 1 shows U_{eff} for $\mu = 0.1$ (an unrealistically heavy M_2, chosen so the five points separate on the page) with its five stationary points.

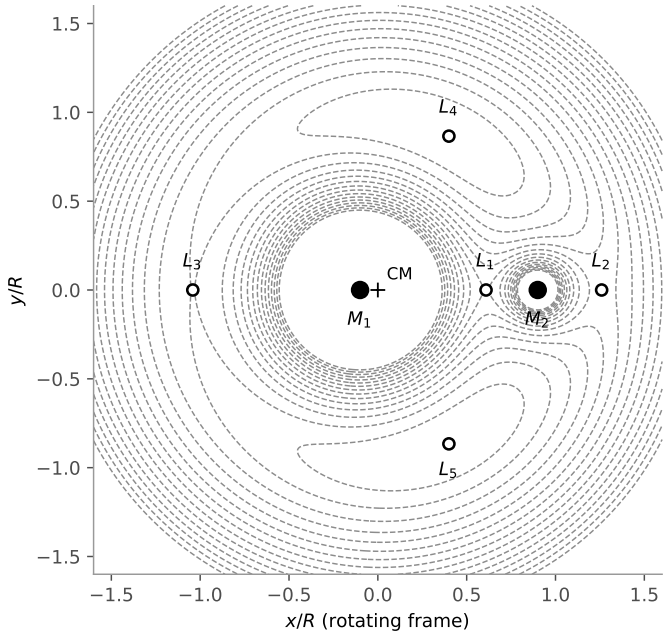


Figure 1: Contours of the effective potential U_{eff} in the rotating frame for $\mu = 0.1$, with the two primaries (filled) and the five Lagrange points (open). L_1, L_2, L_3 are saddle points; L_4 and L_5 are the two maxima. Drawn by `make_figure.py` in the handout's folder.

Board problem (a): L_4 and L_5. Show that the two points forming equilateral triangles with M_1 and M_2 satisfy (2.8), whatever the mass ratio. Work with the forces per unit mass, and use (2.1) and the fact that the CM is at the origin.

Solution. Let $\mathbf{r}$ be such a point and $\mathbf{r}_1$, $\mathbf{r}_2$ the primaries. By construction $|\mathbf{r} - \mathbf{r}_1| = |\mathbf{r} - \mathbf{r}_2| = R$,

so gravity per unit mass is

$$-\frac{1}{m}\boldsymbol{\nabla} U = \frac{GM_1(\,\mathbf{r}_1 - \mathbf{r}\,)}{R^3} + \frac{GM_2(\,\mathbf{r}_2 - \mathbf{r}\,)}{R^3} = \frac{G}{R^3}\Big[(M_1\,\mathbf{r}_1 + M_2\,\mathbf{r}_2) - (M_1+M_2)\,\mathbf{r}\Big]$$
$$= -\frac{G(M_1+M_2)}{R^3}\,\mathbf{r} = -\Omega^2\,\mathbf{r}\,, \tag{2.9}$$

because $M_1\,\mathbf{r}_1 + M_2\,\mathbf{r}_2 = (M_1+M_2)\,\mathbf{r}_{\rm cm} = 0$. The centrifugal force per unit mass is $+\Omega^2\,\mathbf{r}$. They cancel. In words: at the equilateral points the two bodies' gravity points exactly at the CM, with exactly the strength a circular orbit of angular speed Ω needs at that radius, so the test mass goes around the CM in step with the primaries. The dashed triangle in the diagram of the setup is this point. □

Board problem (b): L_1 and L_2 for a small μ. Use units with $R = 1$ and $G(M_1+M_2) = 1$, so $\Omega = 1$ by (2.1), $M_1 \to 1-\mu$ at $x = -\mu$ and $M_2 \to \mu$ at $x = 1-\mu$. On the axis $y = 0$ the condition (2.8) is

$$x = \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} + \frac{\mu\,(x-1+\mu)}{|x-1+\mu|^3}. \tag{2.10}$$

(Check the left side: $-\partial U_{\rm eff}/\partial x$ per unit mass, moved across.) Show that for $\mu \ll 1$ both L_1 and L_2 sit a distance

$$d \approx R\left(\frac{\mu}{3}\right)^{1/3} \tag{2.11}$$

from M_2, one on each side, and evaluate it for the Sun and the Earth ($\mu = 3.00\times10^{-6}$, $R = 1.496\times10^{8}$ km). This d is called the *Hill radius*; inside it M_2's gravity wins over the tidal pull of M_1, which is why the Moon is Earth's.

Solution. Take L_2 first: $x = 1-\mu+\delta$ with $\delta > 0$ (beyond M_2). Keep first order in δ, drop μ wherever it is added to a number of order one, but keep μ/δ^2, which is the term that matters. The left side of (2.10) is $1+\delta$; the first term on the right is $(1-\mu)/(1+\delta)^2 \approx 1-2\delta$; the second is $\mu\delta/\delta^3 = \mu/\delta^2$. So

$$1+\delta \approx 1-2\delta+\frac{\mu}{\delta^2} \quad\Rightarrow\quad 3\delta = \frac{\mu}{\delta^2} \quad\Rightarrow\quad \delta^3 = \frac{\mu}{3}. \tag{2.12}$$

For L_1 put $x = 1-\mu-\delta$: the left side is $1-\delta$, the first term is $(1-\mu)/(1-\delta)^2 \approx 1+2\delta$, and the second is $\mu(-\delta)/\delta^3 = -\mu/\delta^2$, giving $-3\delta = -\mu/\delta^2$ and the same δ. To this order L_1 and L_2 are mirror images about M_2. For the Sun and the Earth, $(\mu/3)^{1/3} = (1.00\times10^{-6})^{1/3} = 0.0100$, so $d = 0.0100$ AU $= 1.50\times10^{6}$ km, about four times the Moon's distance. JWST orbits around L_2 (a "halo orbit" a few hundred thousand km across), so that it always has the Sun, the Earth and the Moon on the same side, behind its sunshield. □

The exact positions. Equation (2.10) is a quintic once the absolute values are resolved, so the three collinear points are found numerically. Solving it with `scipy.optimize.brentq` on the three intervals (left of M_1, between, right of M_2):

	μ	L_1: inside M_2	L_2: beyond M_2	d from (2.11)	L_3, x/R
Sun–Earth	3.00×10^{-6}	1.49×10^{6} km	1.50×10^{6} km	1.50×10^{6} km	-1.000001
Sun–Jupiter	9.54×10^{-4}	5.19×10^{7} km	5.43×10^{7} km	5.31×10^{7} km	-1.0004
Earth–Moon	1.22×10^{-2}	5.80×10^{4} km	6.45×10^{4} km	6.13×10^{4} km	-1.005

The approximation (2.11) is good to a tenth of a percent for the Earth and to about five percent for the Moon, whose μ is not so small. L_3 sits on the far side of M_1, essentially on M_2's orbit. The code is a few lines:

```
from scipy.optimize import brentq
def f(x, mu):
    return x - (1-mu)*(x+mu)/abs(x+mu)**3 - mu*(x-1+mu)/abs(x-1+mu)**3
mu = 3.003e-6
L3 = brentq(f, -2.0, -mu-1e-9, args=(mu,))
L1 = brentq(f, -mu+1e-9, 1-mu-1e-9, args=(mu,))
L2 = brentq(f, 1-mu+1e-9, 2.5, args=(mu,))
```

Noether again: the Jacobi integral. $\mathcal{L}$ in (2.3) has no explicit t, so Section 7.8 says $\mathcal{H} = p_x\dot{x} + p_y\dot{y} - \mathcal{L}$ is conserved. With $p_x = m(\dot{x} - \Omega y)$ and $p_y = m(\dot{y} + \Omega x)$,

$$\mathcal{H} = m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) - \mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + U_{\text{eff}}(x, y). \tag{2.13}$$

This is the Jacobi integral (celestial mechanics usually quotes $C_J = -2\mathcal{H}/m$). Three remarks, the first two of which are the point of Section 7.8:

(a) It is conserved, and it looks like an energy: kinetic energy in the rotating frame plus the effective potential.

(b) It is *not* $T + U$. The true kinetic energy is $\tfrac{1}{2}m(\dot{X}^2 + \dot{Y}^2)$, and from Step 1

$$T + U = \mathcal{H} + m\Omega(x\dot{y} - y\dot{x}) + m\Omega^2(x^2 + y^2), \tag{2.14}$$

which is not constant. Taylor's condition for $\mathcal{H} = T + U$ was that the relation between the generalized coordinates and the inertial position not involve t; (2.2) involves t, so the condition fails, exactly as for the spinning hoop. (And $T+U$ is not conserved in the inertial frame either: there the primaries move and U depends on time. The symmetry lives in the rotating frame, and so does the conservation law.)

(c) Since $\tfrac{1}{2}mv_{\text{rot}}^2 \geq 0$, a body with a given $\mathcal{H}$ can only be where $U_{\text{eff}} \leq \mathcal{H}$. The contours in Figure 1 are therefore fences (Hill's *zero-velocity curves*): a satellite with $\mathcal{H}$ below $U_{\text{eff}}(L_1)$ that starts near M_2 can never leave M_2's neighbourhood, however it moves. Raising $\mathcal{H}$ past $U_{\text{eff}}(L_1)$ opens a gate to M_1; past $U_{\text{eff}}(L_2)$, a gate to the outside.

Stability, in one paragraph (not derived today). L_1, L_2 and L_3 are saddle points of U_{eff} and are unstable: JWST fires its thrusters every few weeks to stay near L_2. L_4 and L_5 are *maxima* of U_{eff}, and yet they are stable whenever $27\mu(1-\mu) < 1$, i.e. $\mu < 0.0385$, which covers Sun–Jupiter (the Trojans) and Earth–Moon. A stable equilibrium at a potential maximum is impossible for a static potential: the small-oscillation analysis of Taylor's Example 7.7 would call it unstable. What rescues it is the Coriolis force, which does no work but keeps turning the velocity, so a body that starts to fall off the hill is swung around it instead. Showing that means linearizing (2.7) *with* the velocity terms kept, a 4×4 eigenvalue problem; it is a very good project, and the same 4×4 problem comes back in Chapter 9.